A decorative graphic on the left side of the slide, consisting of a black crosshair overlaid on a blue square, a red square, and a yellow square.

Lecture 4

Chapter 1. The Foundations

1.3 Predicates and Quantifiers

Previously...

- In predicate logic, a **predicate** is modeled as a **propositional function $P(\cdot)$** from subjects to propositions.
 - $P(x)$: “ x is a prime number” (x : any subject)
 - $P(3)$: “3 is a prime number.” (proposition!)
- Propositional functions of **any** number of arguments, each of which may take **any** grammatical role that a noun can take
 - $P(x,y,z)$: “ x gave y the grade z ”
 - $P(\text{Mike}, \text{Mary}, A)$: “**Mike** gave **Mary** the grade **A**.”

Universe of Discourse (U.D.)

- The power of distinguishing subjects from predicates is that it lets you state things about *many* objects at once.
- e.g., let $P(x) = "x + 1 > x"$. We can then say, "For **any** number x , $P(x)$ is true" instead of $(0 + 1 > 0) \wedge (1 + 1 > 1) \wedge (2 + 1 > 2) \wedge \dots$
- The collection of values that a variable x can take is called x 's ***universe of discourse*** or the ***domain of discourse*** (often just referred to as the ***domain***).

Quantifier Expressions

- **Quantifiers** provide a notation that allows us to *quantify (count) how many* objects in the universe of discourse satisfy the given predicate.
- “ $\forall$ ” is the FORALL or **universal** quantifier.
 $\forall x P(x)$ means for all x in the domain, $P(x)$.
- “ $\exists$ ” is the EXISTS or **existential** quantifier.
 $\exists x P(x)$ means there exists an x in the domain (that is, 1 or more) such that $P(x)$.

The Universal Quantifier $\forall$

- $\forall x P(x)$: *For all x in the domain, $P(x)$.*
- $\forall x P(x)$ is
 - *true* if $P(x)$ is true for every x in D (D : domain of discourse)
 - *false* if $P(x)$ is false for at least one x in D
 - For every real number x , $x^2 \geq 0$ **TRUE**
 - For every real number x , $x^2 - 1 > 0$ **FALSE**
- A **counterexample** to the statement $\forall x P(x)$ is a value x in the domain D that makes $P(x)$ false
- What is the truth value of $\forall x P(x)$ when the domain is empty? **TRUE**

The Universal Quantifier $\forall$

- If all the elements in the domain can be listed as $x_1, x_2, \dots, x_n$ then, $\forall x P(x)$ is the same as the conjunction:

$$P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$$

- Example: Let the domain of x be parking spaces at UH. Let $P(x)$ be the statement “ x is full.” Then the ***universal quantification*** of $P(x)$, $\forall x P(x)$, is the *proposition*:
 - “All parking spaces at UH are full.”
 - or “Every parking space at UH is full.”
 - or “For each parking space at UH, that space is full.”

The Existential Quantifier $\exists$

- $\exists x P(x)$: *There exists an x in the domain (that is, 1 or more) such that $P(x)$.*
- $\exists x P(x)$ is
 - *true* if $P(x)$ is true for at least one x in the domain
 - *false* if $P(x)$ is false for every x in the domain
- What is the truth value of $\exists x P(x)$ when the domain is empty? FALSE
- If all the elements in the domain can be listed as $x_1, x_2, \dots, x_n$ then, $\exists x P(x)$ is the same as the disjunction:

$$P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$$

The Existential Quantifier $\exists$

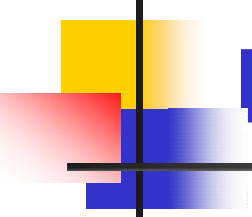
- Example:

Let the domain of x be parking spaces at UH.

Let $P(x)$ be the statement “ x is full.”

Then the ***existential quantification*** of $P(x)$, $\exists x P(x)$, is the *proposition*:

- “Some parking spaces at UH are full.”
- or “There is a parking space at UH that is full.”
- or “At least one parking space at UH is full.”

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Free and Bound Variables

- An expression like $P(x)$ is said to have a **free variable** x (meaning, x is undefined).
- A quantifier (either $\forall$ or $\exists$) *operates* on an expression having one or more free variables, and **binds** one or more of those variables, to produce an expression having one or more **bound variables**.

Example of Binding

- $P(x,y)$ has 2 free variables, x and y .
- $\forall x P(x,y)$ has 1 free variable y , and one bound variable x . [Which is which?]
- “ $P(x)$, where $x = 3$ ” is another way to bind x .
- An expression with zero free variables is a bona-fide (actual) proposition.
- An expression with one or more free variables is not a proposition:

e.g. $\forall x P(x,y) = Q(y)$

Quantifiers with Restricted Domain

- Sometimes the universe of discourse is restricted within the quantification, e.g.,
 - $\forall x > 0 P(x)$ is shorthand for
“For all x that are greater than zero, $P(x)$.”
 $= \forall x (x > 0 \rightarrow P(x))$
 - $\exists x > 0 P(x)$ is shorthand for
“There is an x greater than zero such that $P(x)$.”
 $= \exists x (x > 0 \wedge P(x))$

Translating from English

- Express the statement “*Every student in this class has studied calculus*” using predicates and quantifiers.
 - Let $C(x)$ be the statement: “*x has studied calculus.*”
 - If domain for x consists of the students in this class, then
 - it can be translated as $\forall x C(x)$

or

- If domain for x consists of all people
- Let $S(x)$ be the predicate: “*x is in this class*”
- Translation: $\forall x (S(x) \rightarrow C(x))$

Translating from English

- Express the statement “*Some students in this class has visited Mexico*” using predicates and quantifiers.
 - Let $M(x)$ be the statement: “ x has visited Mexico”
 - If domain for x consists of the students in this class, then
 - it can be translated as $\exists x M(x)$
 - or
 - If domain for x consists of all people
 - Let $S(x)$ be the statement: “ x is in this class”
 - Then, the translation is $\exists x (S(x) \wedge M(x))$

Translating from English

- Express the statement “*Every student in this class has visited either Canada or Mexico*” using predicates and quantifiers.
- Let $C(x)$ be the statement: “ x has visited Canada” and $M(x)$ be the statement: “ x has visited Mexico”
- If domain for x consists of the students in this class, then
- it can be translated as $\forall x (C(x) \vee M(x))$

Negations of Quantifiers

- $\forall x P(x)$: “Every student in the class has taken a course in calculus” ($P(x)$: “ x has taken a course in calculus”)
 - “Not every student in the class ... calculus”
 $\neg \forall x P(x) \equiv \exists x \neg P(x)$
- Consider $\exists x P(x)$: “There is a student in the class who has taken a course in calculus”
 - “There is no student in the class who has taken a course in calculus”
 $\neg \exists x P(x) \equiv \forall x \neg P(x)$

Negations of Quantifiers

- Definitions of quantifiers: If the domain = $\{a, b, c, \dots\}$
 - $\forall x P(x) \equiv P(a) \wedge P(b) \wedge P(c) \wedge \dots$
 - $\exists x P(x) \equiv P(a) \vee P(b) \vee P(c) \vee \dots$
- From those, we can prove the laws:
 - $\neg \forall x P(x) \equiv \neg(P(a) \wedge P(b) \wedge P(c) \wedge \dots)$
 $\equiv \neg P(a) \vee \neg P(b) \vee \neg P(c) \vee \dots$
 $\equiv \exists x \neg P(x)$
 - $\neg \exists x P(x) \equiv \neg(P(a) \vee P(b) \vee P(c) \vee \dots)$
 $\equiv \neg P(a) \wedge \neg P(b) \wedge \neg P(c) \wedge \dots$
 $\equiv \forall x \neg P(x)$
- Which *propositional* equivalence law was used to prove this?

DeMorgan's

Negations of Quantifiers

Theorem:

- **Generalized De Morgan's laws for logic**

1. $\neg \forall x P(x) \equiv \exists x \neg P(x)$

2. $\neg \exists x P(x) \equiv \forall x \neg P(x)$

Negations: Examples

- What are the negations of the statements $\forall x (x^2 > x)$ and $\exists x (x^2 = 2)$?
 - $\neg \forall x (x^2 > x) \equiv \exists x \neg (x^2 > x) \equiv \exists x (x^2 \leq x)$
 - $\neg \exists x (x^2 = 2) \equiv \forall x \neg (x^2 = 2) \equiv \forall x (x^2 \neq 2)$
- Show that $\neg \forall x (P(x) \rightarrow Q(x))$ and $\exists x (P(x) \wedge \neg Q(x))$ are logically equivalent.
 - $$\begin{aligned} \neg \forall x (P(x) \rightarrow Q(x)) &\equiv \exists x \neg (P(x) \rightarrow Q(x)) \\ &\equiv \exists x \neg (\neg P(x) \vee Q(x)) \\ &\equiv \exists x (P(x) \wedge \neg Q(x)) \end{aligned}$$

Summary

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TABLE 1 Quantifiers.

<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall x P(x)$	$P(x)$ is true for every x .	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x .

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TABLE 2 De Morgan's Laws for Quantifiers.

<i>Negation</i>	<i>Equivalent Statement</i>	<i>When Is Negation True?</i>	<i>When False?</i>
$\neg \exists x P(x)$	$\forall x \neg P(x)$	For every x , $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg \forall x P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x .